VF Asia Footwear Laboratory Testing Manual

Test Method for Water Swelling	Test method # VF- E18 Page 1 of 2
Date Issued : May 27, 2008 Revision # :	Created by : Nila Babia Erazo/ Chris Adam

PURPOSE AND SCOPE: To measure the swelling of the test material as a result of being immersed in water for a specified period of time.

REFERENCE: ASTM D 1815

EQUIPMENT AND MATERIAL:

1. Immersion dish which is large enough to contain the test sample such that no portion of the test sample contacts the sides of the dish and the test sample is completely immersed in the water.
2. Blotting paper
3. Piece of glass rod for supporting the test material while being immersed in water.
4. Vernier Caliper capable of measuring to +/-0.1cm.
5. Insole Board, Foam

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at standard laboratory environment at 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.
2. Die or hand cut samples which are at least 12.0cm long and have a width of 2.54cm.
3. Draw a line (L) which is 10.0cm long along each of the test samples .line L should be drawn one cm from each edge of the test sample.

REPETITIONS: Test two (2) samples per material**.**

TEST PREOCEDURE:

1. Using a Vernier caliper, measure the length of line L to the nearest 0.1mm (L_1).
2. Place the test sample in the immersion dish, support the test samples by placing a glass rod underneath each end of the test sample.
4. Carefully pour distilled water at 23+/-2°C into the immersion dish until the test sample is completely immersed, use at least 10 grams of water for every gram which the test sample weighs
5. Allow the test sample to remain immersed in water for 120+/-5 minutes
6. Remove the test sample from the water ,and lightly blot the surface of the sample to remove any surface water
7. Immediately check the length of the material to the nearest to +/-0.1cm. The final length is L2.

Test Method for Water Swelling	Test method # VF- E18 Page 2 of 2
Date Issued : May 27, 2008 Revision # :	Created by : Nila Babia Erazo/ Chris Adam

CALCULATIONL

Water Swelling (%): = $\frac{L2-L1}{L1}$ *100

REPORT: The average of the two measurements to the nearest 0.1%.